

SEMESTER-III

UME517: MATERIALS ENGINEERING AND METALLURGY				
	L	T	P	Cr
	2	1	0	2.5
Course Objective: This course identifies the key variables and strengthening mechanisms which affect the mechanical properties of engineering materials. It describes the correlation between internal structure of materials and their mechanical properties. The course provides a detailed interpretation of phase diagrams. It describes the role of Fe-Fe₃C diagram, TTT diagrams, and CCT diagrams in tailoring the microstructure and hence mechanical properties of steels under various processing conditions. It explains the kinetics of formation and decomposition of austenite in iron-carbon alloys. The course describes the various heat treatment and case hardening processes for industrial processing of materials. It provides an overview to other mechanical engineering materials like stainless steels, titanium and nickel based alloys.				
Syllabus Basics of Materials Engineering: Mechanical properties and their evaluation, Crystal structure and imperfections in solids, Strengthening mechanisms, Fundamental terms in materials engineering and their significance viz. phases, components, equilibrium, and degrees of freedom. Gibb's phase rule and Inverse lever rule. Phase Diagrams: Significance of phase diagrams, Classification of phases in alloys based on mechanical properties and also on basis of equilibrium, Equilibrium diagrams for isomorphous and eutectic alloy systems, Iron-cementite equilibrium diagram, Invariant reactions of iron-carbon systems, Critical temperature lines, Transformations and microstructure evolution in iron-carbon alloys. Kinetics of Austenite Transformations: Kinetics of formation and decomposition of austenite in steels, Austenite grain size, Effect of microstructure on various mechanical properties, Time Temperature Transformation diagrams (TTT Diagrams), Continuous cooling transformation diagrams (CCT diagrams). Heat Treatment and Case Hardening Treatment: Procedural details, microstructural changes and mechanical property improvements obtained through various types of annealing, normalizing, hardening and tempering heat treatments. General features of surface hardening processes and chemical heat treatment processes (carburizing, nitriding, and cyaniding, flame hardening, induction hardening etc.) Advanced Materials: Alloying of steels, Stainless steels, Maraging steels-cast irons, Titanium and nickel based alloys.				
Course Learning Outcomes (CLOs) The students will be able to: 1. develop and critically analyze the phase diagrams of isomorphous alloys, eutectic series alloys, and iron-carbon alloys 2. analyze the microstructures that exist in a given engineering material for describing structure-mechanical property relationship 3. obtain and interpret information from the TTT curves and CCT diagrams of different alloys				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.

4. select and analyze suitable heat treatment/case hardening process for improvement in specific mechanical properties

Text Books

1. Higgins, R. A., Engineering Metallurgy-Applied Physical Metallurgy, Elsevier (2012).
2. W. D. Callister, Materials Science and Engineering - An Introduction, 6th Edition, Wiley India (2017).

Reference Books

1. Rajan, T.V., Sharma, C. P., Sharma, A., Heat Treatment Principles and Techniques, PHI Learning Private Limited (2011).
2. Abbaschian, R. and Reed-Hill, R., Physical Metallurgy Principles, Cengage Learning (2008).
3. George E. Dieter, Mechanical Metallurgy, McGraw-Hill Education (2017)

Evaluation Scheme:

S. No.	Evaluation Elements	Weightage (%)
1.	MST	35
2.	EST	45
3.	Sessional (May include tutorial assignments/quizzes/lab evaluation)	20

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.